Nomenclature

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DATE PLACEHOLDER

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1 Alkane Nomenclature

Alkanes are the simplest form of hydrocarbons—compounds composed entirely of C and H atoms in a linear, single-bond chain. Their names are based on the number of carbon atoms in the chain. For instance, the simplest form is methane (CH_4), then ethane (C_2H_6), and etc. The chemical formula of an alkanes follows a pattern and is expressed as C_nH_{2n+2}.

Carbon Count	Alkane Name
1	Methane
2	Ethane
3	Propane
4	Butane
5	Pentane
6	Hexane
7	Heptane
8	Octane
9	Nonane
10	Decane
11	Undecane
12	Dodecane
13	Tridecane
14	Tetradecane

An alkyl substituent is essentially its cooresponding alkane with one H removed at the end. Its chemical formula is expressed as $C_{n-1}H_{2n+1}$, and its naming scheme is identical to that of alkanes except that the end ”-ane” is replaced with ”-yl” (e.g. ethyl (C_2H_5)). Propyl is shown as an example below.

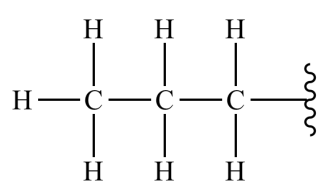

Figure 1: Propyl Substituent

When naming organic molecules, chemists follow a standardized nomenclature system made by the **International Union of Pure and Applied Chemistry (IUPAC)**. The process of naming an organic molecule is done in the following steps.

(1) Find the longest alkane chain (stem) and name it based on its number of carbon atoms as we have done above. When there are two **equal longest** chains, choose the one with **more substituents**.

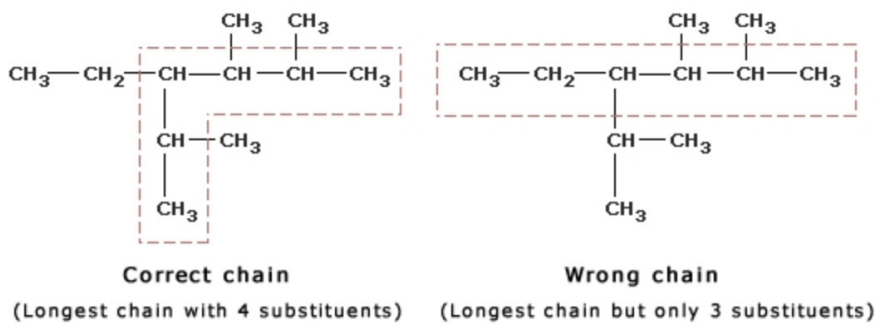

Figure 2: Longest Chain Selection

In the example above, the longest chain has 6 carbons, thus we name it **hexane**. Recognize that the left chain has more substituents than the right chain, thus the left chain is chosen in the context of naming.

(2) Name the substituents as either **alkyl** (ethyl, propyl, butyl, etc.) or **halo** (bromo, fluoro, chloro, iodo). We will refer to substituents as *R* in this section.

(2a) For a **straight chain R**, we simply name the R as we have defined in (2).

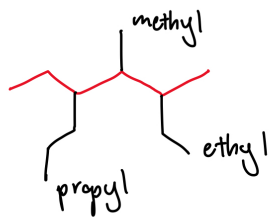

Figure 3: Straight Chain Substituent

(2B) For a **branched chain R**, we again find the longest chain within R and also name the substituents of this chain within R.

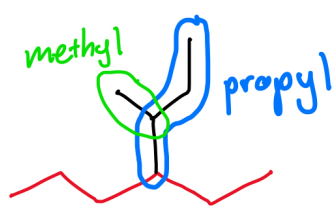

Figure 4: Branched Chain Substituent

The branched R group above would be named methylpropyl.

(2c) When we face multiple of the same substituents, we simply use the prefixes di-, tri-, tetra-, etc. for straight chain R groups and bis-, tris-, tetrakis-, etc. for branched chain R groups.

Here is an example of naming multiple straight chain R groups.

Figure 5: Multiple Straight Chain Substituents

Here is an example of naming multiple branched chain R groups.

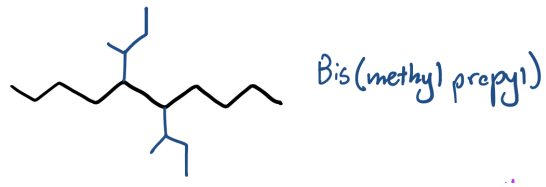

Figure 6: Multiple Branched Chain Substituents

(2d) Although there is a systematic name for every type of alkyl substituent, there are some colloquially used names for certain alkyl substituents that can only be learned through memorization.

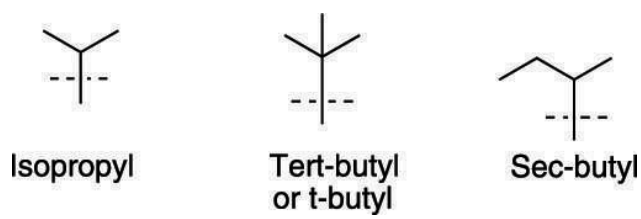

Figure 7: Common Alkyl Substituent Names

(3) Now we have to number the carbons along the stem alkane in order to express the positions of the substituents. The numbering scheme starts from the end that is **closest to a substituent**.

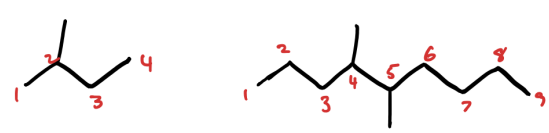

Figure 8: Numbering Closest to Substituent

It both ends are equidistant to the first substituent, proceed until the **first point of difference**.

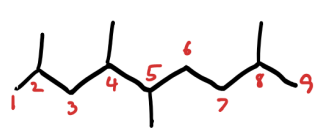

Figure 9: First Point of Difference Rule

And if the positioning of the substituents are exactly the same, the numbering begins in the end where the first substituent's name is **alphabetically first**.

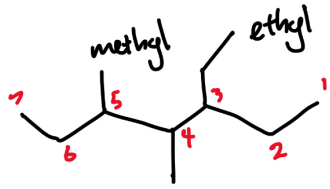

Figure 10: Alphabetical Numbering Rule

As you can see above, the positioning of the substituents are identical from both ends, however ethyl comes alphabetically first relative to methyl, thus the numbering starts from the right end.

This same numbering scheme is used in numbering the stem of a branched chain R group. However, we always start from the carbon of attachment in this case.

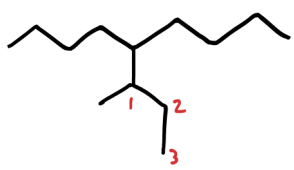

Figure 11: Branched Chain Numbering

(4) Now that we have determined the names and positions of the substituents, we can name the alkane. Keep in mind that we order the substituent names **alphabetically** (not numerically). The locations are given by a number prefix coorespondingly.

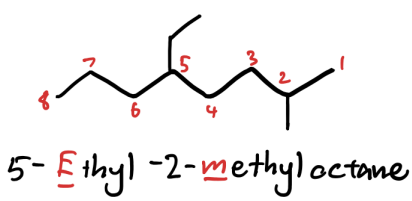

Figure 12: Alphabetical Ordering of Substituents

Also keep in mind that the alphabetical order contribution of prefixes di-, tri-, bis-, etc. are **not counted** for the main stem R groups, but **are counted** when in branched R groups.

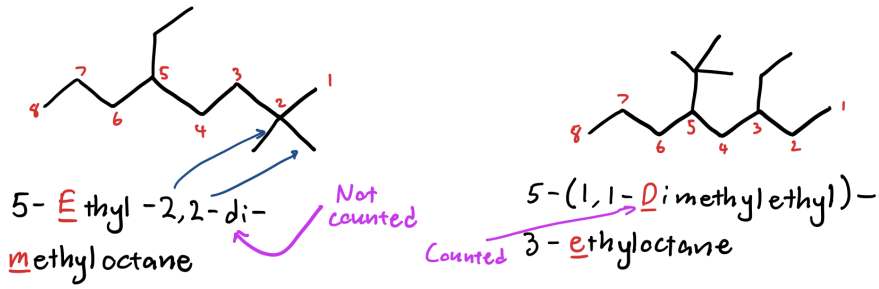

Figure 13: Alphabetical Counting of Prefixes

Let's take a look at one final large example.

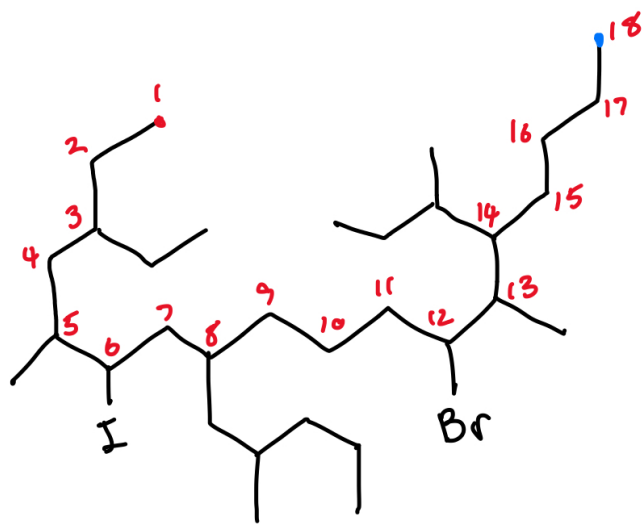

Figure 14: Complex Nomenclature Example

The name of the compound above would be the following:

12-bromo-3-ethyl-6-iodo-5,13-dimethyl-8-(2-methylpentyl)-14-(2-methylpropyl)octadecane

2 Cycloalkane Nomenclature

Cycloalkanes are alkane rings that are named according to the number of carbons with a "cyclo-" prefix.

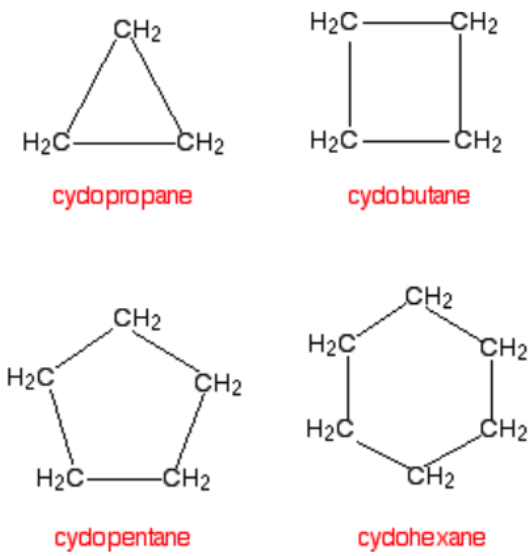

Figure 15: Cycloalkane Examples

Within a compound that contains both a cycloalkane and a chain alkane, the stem alkane is still chosen as the one with the most carbons.

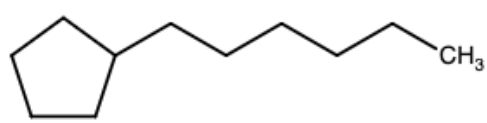

Figure 16: Cycloalkane as Substituent

For instance the compound above would be called **1-cyclopentylhexane**.

Under the case of a cycloalkane having more carbons than the chain alkane, the stem is the cycloalkane.

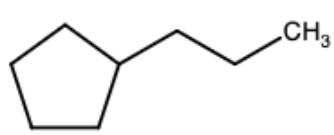

Figure 17: Chain Alkane as Substituent

The compound above would be called **propylcyclopentane**. Notice that we do not need a number prefix for the propyl because there is only one substituent to the stem cycloalkane and the numbering of cycloalkane carbons would be arbitrary in this case.

When there are multiple substituents to the cycloalkane, we (1) number the carbons in a way that requires the lowest digit sequence and (2) name the substituents alphabetically in the compound name.

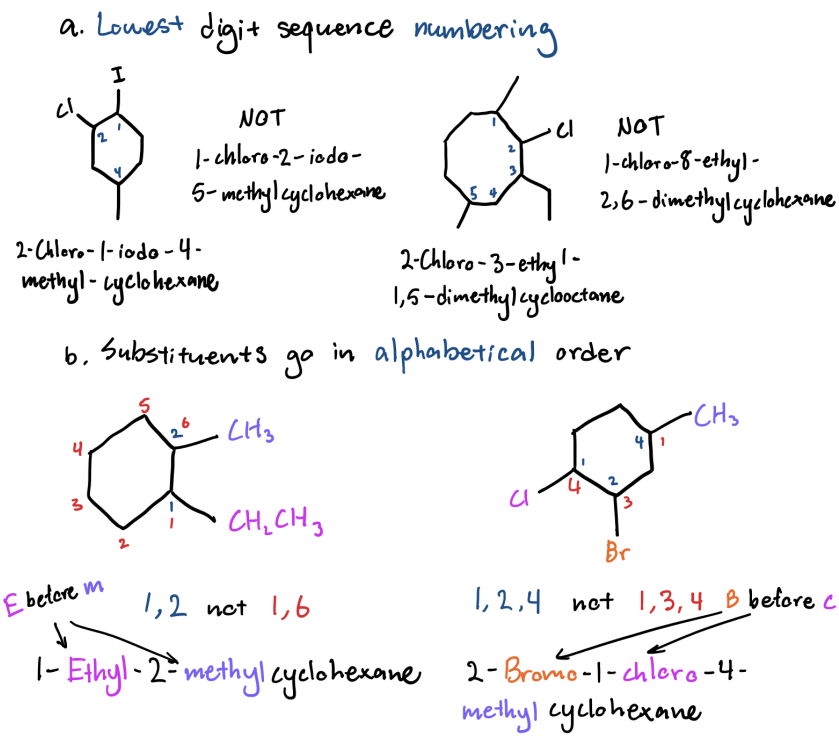

Figure 18: Multiple Substituents on Cycloalkane

In a scenario where multiple cycloalkanes are present in a compound, the smaller unit is substituent to the larger. Below is an example of **cyclopentylcyclohexane**.

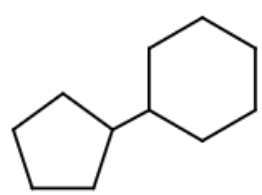

Figure 19: Nested Cycloalkanes

Now we introduce **stereoisomers**, which are molecules that have the same molecular formula and the same sequence of bonded atoms (constitution), but differ in the 3D orientation of those atoms in space. In cycloalkanes, we have two sides: "up" and "down". With two (or more) substituents, we define substituents bonding towards the same side as **cis** and opposite sides as **trans**.

cis-
1-chloro-2-methylcyclohexane

trans-
1-chloro-2-methylcyclohexane

Figure 20: Cis and Trans Isomers

Pointing away from you (below the plane of the ring) is represented by **dashes** and pointing towards you (above the plane of the ring) is represented by **wedges** as shown above.

3 Absolute Configuration: The Gold Standard

While the cis and trans system is convienent and sufficient in the context of naming simple cycloalkanes, there are more complex cases that require us to use another naming convention: **absolute configuration**.

The **absolute configuration** system is a universal language that allows us to describe the orientation of a substituent of either a chain alkane or a cycloalkane in the context of the carbon it is attached to (a.k.a. its **chiral center**)

A **chiral center** is usually a carbon atom that has four different groups attached to it in a tetrahedral manner.

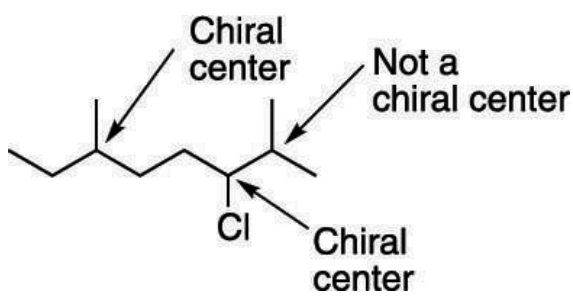

Figure 21: Chiral Center Examples

As you can see above, the chiral centers have four distinct groups attached to it defined as the following:

Left chiral center → ethyl, methyl, hydrogen, *rest of the molecule*

Right chiral center → Chloro, isopropyl, hydrogen, *rest of the molecule*

On the other hand, the non-chiral center is not chiral because it has two methyl groups attached to it simultaneously.

Non-chiral center → methyl, methyl, hydrogen, *rest of the molecule*

Once we have defined the chiral center, we can now move onto ranking the order of the attached groups based on the atomic mass of the adjacent atom.

Let's focus in on the left chiral center. it is attached to three carbons and one hydrogen. Since we know that hydrogen is lighter than carbon, we can already rank the hydrogen atom at the bottom (or a rank of 4). Next, when we are given a scenario where the adjacent atoms are identical, we simply go further down the chain until we reach a **point of difference**.

Ethyl → C, H, H

Methyl → H, H, H

Rest of the molecule → C, H, H

Based on the above assessment of the atoms attached to the adjacent carbons, we can now rank the methyl group as 3.

Skipping the rest of the explicit explanation, we can see that the rest of the molecule (ROM) would rank 1 and the ethyl would rank 2.

[1] ROM

[2] Ethyl

[3] Methyl

[4] Hydrogen

R and S is defined as whether the ordering of the rank 1, 2, and 3 groups is **clockwise** (R) or **counterclockwise** (S) when **looking at the chiral center with the rank 4 group sitting exactly behind it**.

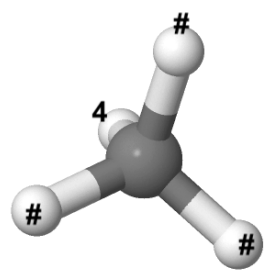

Figure 22: Absolute Configuration Perspective

Now, since the original figure does not actually define the orientation (wedge or dash) of the methyl group relative to the stem alkane, we cannot actually define its R/S orientation. Thus let's go through both potential cases.

First, the wedge case:

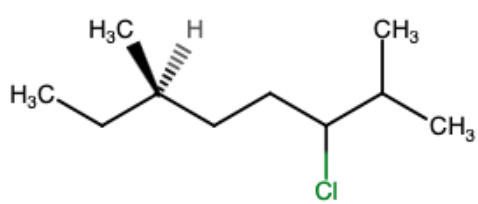

Figure 23: Chiral Center with Wedge

Here, the molecule is already oriented in a manner that shows it already from the angle we must analyze it from because the rank 4 group (hydrogen) sits behind the chiral center. the arrows from the rank 1 (ROM) to the rank 2 (Ethyl) to the rank 3 (methyl) follow a clockwise path, thus we define this chiral center as R.

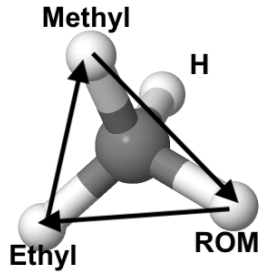

Figure 24: R Configuration

When we move to the dash case,

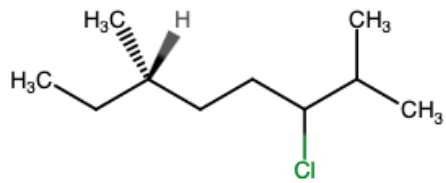

Figure 25: Chiral Center with Dash

we will find the orientation to be counterclockwise following the same procedure as above, thus we define this case of the chiral center as S.

Within the context of nomenclature, the R/S is written in parentheses at the beginning of the molecule name alongside the cooresponding carbon number that represents its chiral center.

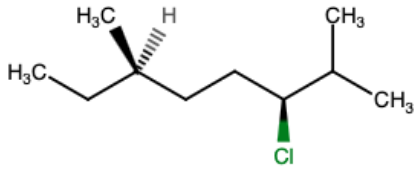

Figure 26: Absolute Configuration in Complex Molecule

Combining all of these rules, we can define the compound shown above as

(3S,6R)-3-chloro-2,6-dimethyloctane

The same process can be applied to cycloalkanes as well.

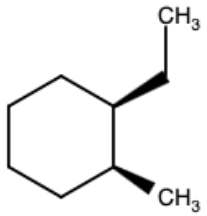

Figure 27: Absolute Configuration in Cycloalkanes

The name of the cycloalkane above is:

(1R,2S)-1-ethyl-2-methylcyclohexane